https://github.com/alhadi-01/DevOps-Prod-Application

Application Deploymet with CI/CD Jenkins on Kubernetes AWS EKS using ArgoCd.

- Source code and configuration is managed at GitHub for code, dependencies, application manifests and respective Jenkins file with stages and Docker file.
- Using GitHub for source code CI/CD pipeline implemented in two stages Continuous Integration with Jenkins and Continuous Deployment using Argocd on Kubernetes or AWS EKS.

- Continuous integration involves :
  - Source code checkout at GitHub repository
  - Application Build using Maven
  - Static Code analysis using SonarQube for code scanning (Note: SonarQube Server to be configured and credentials and need to be added in Jenkins)
  - Using Docker for image build for application and pushing it to Docker-hub maintained repository.
    (Docker credentials need to be configured with Jenkins including GitHub credentials)

- Continuous Deployment using ArgoCD:
  - ArgoCd operator Installed at AWS EKS
  - Tracks the application manifest at GitHub
  - Any commit happens at repository ArgoCd track the changes
  - Updated application manifest deployed on AWS EKS using ArgoCd.

AWS Infrastructure as a code setup with terraform

- Refer GitHub repository for EC2, VPC, EBS, S3 configuration for infra provisioning using Terraform.
- Configuration can be implemented using modules or variables and values depends on the requirement).

Basic Commands to provision infra with respective configurations:

```
terraform init
terraform validate
terraform plan
terraform apply
```

Monitoring and observability using AWS Prometheus and Grafana.

AWS Prometheus and Grafana for setting up monitoring and observability on AWS EKS cluster level.

- For AWS Prometheus Amp workspace creation for data managed by aws with respective workspace id and workspace remote write URL for access.
- Enable aws managed scrapper for collecting metrics and configure it for eks cluster.
- Grained access using Iam for user and service access in aws. And.
- Data ingested to amp workspace through scrappers.
- Create Grafana a workspace for observability dashboard
- Integrating Grafana workspace to aws Prometheus workspace by adding data source using AMP endpoint/access point.
- Import dashboard for Kubernetes pod node service metrics.

CloudWatch and can be implemented to monitor over cluster behaviour and metrics and integrated service for monitoring auditing can be done using CloudTrail AWS Config is a better option to go with precise fine grained implementation for project compliance.

Tool/Area	Rating (1–5)	Comments or Example Projects
CI/CD (e.g., GitHub Actions, Jenkins)	3.8	Majorly worked with ci/cd Jenkins.
Docker	4.0	Have worked with Docker for Kubernetes.
Kubernetes	4.0	Working with AWS EKS in current project.
Terraform / Ansible	4.0	Experienced with Terraform.
AWS / GCP / Azure	4.0	Working with AWS Cloud have knowledge about Azure.
Monitoring Tools (Prometheus, Grafana, ELK)	4.0	Working with AWS Prometheus Grafana, CloudWatch and Cloud Trail
Git	3.8	Experienced with GitHub for source code management.